Runtime Re-Sectoring at the Boundary of W_k and W_{k+1}

on the CPU, while the GPUs apply W_k

❶ Sample

Term	W_k	Term	W_k
Z0X1	11	X0Z1	01
Y1	11	Y0	01
X0Z1Z2	00	Z0X1Z2	10
Y0Z2	00	Y1Y2X3	10

a share from every node

❷ Row Solver

X3 X2 X1 Z0Z1

$GF(2)$

x_0 $z_0 \oplus z_1$ z_2 z_3

kernel of W_{k+1}

❸ Sector Balancer

Row Pair	00	01	10	11	Largest
$z_0 \oplus z_1, z_3$	0	0	8	0	8
z_2, z_3	4	0	4	0	4
x_0, z_2	2	2	2	2	2 ✓

keep the pair whose largest sector is smallest

$g = 2$ *rows: 4 sectors for 4 GPUs*

❹ Rows of W_{k+1}

x_0 z_2

Sector	Node
$00'$	0
$01'$	1
$10'$	0
$11'$	1

$d \bmod N$